Bangladesh Army University of Engineering & Technology (BAUET)

Qadirabad, Natore

Faculty of Electrical and Computer Engineering

Department of Computer Science and Engineering

Lecture Plan

Program:	**Computer Science &Engineering (CSE)**
Course Code & Title:	**HUM-3157**(Professional Ethics and Environmental Protection) **GED-1251** (Society, Ethics and Environment)
Semester:	Summer-2024
Level:	3rdYear5thSemester/1st Year 2nd Semester
Credit Hour:	3.00 Hrs./Week
Name & Designation of Teacher:	Professor Mohammed Golam Sarwar Bhuyan Department of Computer Science and Engineering Dean, Faculty of Electrical and Computer Engineering (ECE) Director, Institutional Quality Assurance Cell (IQAC)
Class Hours:	Tuesday: 0915AM-1010AM 1015 AM-1115 AM Wednesday: 0915AM-1010AM (GSB/AY) 1015 AM-1115 AM (GSB/AY)
Consultation Hours:	Sunday:09:00AM-02:00 PM
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Rationale:	This course provides students with an understanding of the impact of technology on society and the environment. The course examines ethical issues governing social, technological and environmental policies, and the impact of their implementation. Students will learn to analyze, question and resolve ethical dilemmas by developing a personal framework for ethical and moral decision-making. They will explore your responsibilities in your personal and professional lives, in the community and in the global environment. Class discussions will include: privacy versus law enforcement, access to information, freedom of speech versus control of content, stem cell research and medical advances in the context of culture, global warming, green energies and sustainable technologies.
Pre-requisite(ifany):	Not Applicable

Course Objectives:

The objectives of this course are to provide for Computer Science and Engineering students with:

1. **Understanding Ethical Theories and Principles:** Students should be able to comprehend and apply major ethical theories (such as utilitarianism, deontology, virtue ethics) and principles (like justice, autonomy, beneficence) to engineering dilemmas specific to computer science.
2. **Recognizing Ethical Issues in Computing**: Enable students to identify and analyze ethical issues that arise in the practice of computer science and engineering, including topics such as privacy, security, intellectual property, and the impact of technology on society.

3. **Developing Ethical Decision-Making Skills:** Equip students with frameworks and methodologies to systematically analyze and resolve ethical dilemmas in their professional practice as computer scientists and engineers.
4. **Promoting Professional Responsibility:** Instill a sense of responsibility and accountability among students towards stakeholders affected by their work, including clients, employers, colleagues, and the broader society.
5. **Exploring Ethical Standards and Codes of Conduct**: Familiarize students with existing codes of ethics and professional conduct relevant to computer science and engineering disciplines, and encourage adherence to these standards in their future careers.

These objectives aim to prepare students not only to understand the ethical dimensions of their work but also to make informed and responsible decisions as future professionals in the field of computer science and engineering.

Course Content:

Engineering Ethics and Codes: Introduction to Ethics, Theories of Ethics, Principles of Engineering Ethics, Employers and employees, inter-professional relationship, Fundamental Canons, NSPE codes, IEEE codes, ACM codes, Institutionalization of ethical conduct, Ethical Dilemmas **Technology and Computer Ethics:** Computer crime and cyber security, Privacy and Confidentiality issue in CSE, Legal framework, Copyright laws, ICT Act, Right to Information, Patents, Ethical challenges for CSE engineers with advancement of modern technology in society, Case studies related ethical issues in ICT. **Engineering Ethics and Professionalism:** Responsible professionals, corporations, The origin of ethical thought, Ethics and the law, Moral reasoning and codes of ethics, ethical decision making strategies, codes of ethics and case studies, moral issues and ethical theories. **Social issues on Environmental Protection and Law:** Sustainable development goal (SDGs), Environment and natural resources, Global environmental problems, environmental ethics, Pollution, Solution and prevention, Environmental policy and protection tools, Waste: problem and solutions, Urban environment, Industrialization, Climate change, Voluntary tools in the environmental protection, impact of environmental laws in solving environmental problems, Environmental legislation and its importance, Environmental ethics and social responsibility, Importance of sustainability assessment tools and institutions before and after laws are adopted.

Text and Ref Books:

1. Introduction to Engineering Ethics by Mike W. Martin and Roland Schinzinger
2. Engineering Ethics: Concepts and Cases" by Charles E. Harris Jr., Michael S. Pritchard, and Michael J. Rabins

***Additional reading material:** Power Point slides & PDF provided by the teacher

Minimum Attendance:	
80% class attendance is mandatory for a student in order to appear at the final examination.	
Student's Responsibilities:	
1.	Students must come to the class prepared for the course material covered in the previous class
2.	They must submit their assignments on time
3.	No late or partial assignments will be acceptable

Grading Policy:			
Numerical Score	**Letter Grade**		**Grade points**
80%andabove	A+	A(Plus)	4.00
75%to >80%	A	A(Regular)	3.75
70%to >75%	A-	A(Minus)	3.50
65%to >70%	B+	B(Plus)	3.25
60%to >65%	B	B(Regular)	3.00
55%to >60%	B-	B(Minus)	2.75
50%to >55%	C+	C(Plus)	2.50
45%to >50%	C	C(Regular)	2.25
40%to >45%	D	D(Pass)	2.00
below 40%	F	Fail	0.00
Incomplete	I	-	-
Withdrawal	W	-	-
Project/Thesis Continuation	X	-	-

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

CO1	**Demonstrate** an understanding of professional ethics, codes of conduct, and moral responsibility in decision-making processes, addressing real-world challenges in the context of computer science and engineering.
CO2	**Analyze** the impact of engineering solutions on societal well-being and apply ethical frameworks to resolve dilemmas involving technology, privacy, and social justice.
CO3	**Evaluate** the environmental consequences of technological advancements and design sustainable solutions by adhering to principles of environmental protection and resource conservation

COs	PO	KP	Bloom's Level	Delivery methods and activities	Assessment Tools
CO1	7	9	Understand/Respond (C2/A2) Apply/Value (C3/A3) Analyze/ Organize (C4/A4)	Interactive Lectures, Demonstration, Individual/Group, Assignment/Project, Group Discussion, Case Studies, Problem-based Learning, Audio/Video	Class Test/ Assignment Mid-Term Assessment/ Exam, Semester Final Examination (Written)
CO2	6,7	7,9	Understand/Respond (C2/A2) Apply/Value (C3/A3) Analyze/ Organize (C4/A4) Evaluate/ Internalize (C5/A5)		
CO3	6	7	Understand/Respond (C2/A2) Apply/Value (C3/A3) Evaluate/ Internalize (C5/A5)		

COs	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
CO1							M					
CO2						M	M					
CO3						M						
Legends: High (H)=3, Medium(M)=2 and Low(L)=1												

Assessment of CO-PO, Bloom's Level

Teaching-learning and Assessment Strategy:
Class Participation/Presentation, Class Performance, Homework/Assignment, Class Test, Mid Term Assessment/Examination (Project/Written), Semester Final Examination (Written)

Assessment Methods	Description	Marks	Cos	Blooms Taxonomy	Remark
Continuous Assessment (20%)	**Class Test/Assignment** **Class Performance**	15% 5%	CO1 CO2 CO3 CO4	Understand/Respond (C2/A2) Apply/Value (C3/A3) Analyze/ Organize (C4/A4)	The best 3 will be counted
Mid Term Assessment/ Examination (20%)	(Project/Written)	20%	CO1 CO2	Understand/Respond (C2/A2) Apply/Value (C3/A3) Analyze/ Organize (C4/A4)	[There will be 03 questions. Student have to answer question No. 1 which is compulsory and another one (1) from question No. 2 or 3]
Semester Final Examination (60%)	(Written)	60%	CO1 CO2 CO3 CO4	Understand/Respond (C2/A2) Apply/Value (C3/A3) Analyze/ Organize (C4/A4) Evaluate/ Internalize (C5/A5)	[There will be two sections namely **A** and **B**. Each section will have 4 questions. Question No. 1 and 2 from section **A** and 5 and 6 from section **B** are compulsory and student have to answer another one (1) from question No. 3 or 4 and 7 or 8]. **Note:** There will be two (2) compulsory questions in the semester final exam. from the mid-term syllabus.
Total Marks		**100%**			

Teaching and Learning Strategies (TLS)	
TLS1	Delivery lectures using multimedia of different topics
TLS2	Active and fruitful discussion in class regarding efficient solving of the logical, mathematical and complex engineering problems
TLS3	Group discussion and presentation regarding diverse problems and corresponding lectures
TLS4	Evaluation of class performance to reach each student in a class for every topic.

Graduate Attributes/Program Outcomes (POs)

Program Outcomes (POs) represent the knowledge, skills and attitudes the students should have at the end of a four years engineering program. CSE program of BAUET has 12 Program Outcomes. They are briefly described in the following table.

PO	Category	Description
PO1	Engineering Knowledge	Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions of complex engineering problems
PO2	Problem Analysis	Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development (WK1 to WK4).
PO3	Design creative solutions	Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate consideration for public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations as required (WK5).
PO4	Investigation	Conduct investigations of complex engineering problems using research methods including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions (WK8).
PO5	Modern Tool Usage	Create, select and apply and recognize limitations of appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering problems (WK2, WK6).
PO6	The Engineer: Society, Economy, Environment and Sustainability	When solving complex engineering problems, analyze and evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (WK1, WK5, and WK7).
PO7	Ethics	Apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws. Demonstrate an understanding of the need for diversity and inclusion (WK9).
PO8	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings (WK9).
PO9	Communication	Communicate effectively and inclusively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, taking into account cultural, language, and learning differences
PO10	Project Management	Apply knowledge and understanding of engineering management

		principles and economic decision-making and apply these to one's own work, as a member and leader in a team and to manage projects and in multidisciplinary environments
PO11	Life Long Learning	Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change (WK8).
PO12	Entrepreneurship	Demonstrate knowledge and understanding of the competences necessary to transform opportunities and ideas into a new business.

Knowledge and Attribute Profile (KPs)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences
WK2	Conceptually based mathematics, numerical analysis, data analysis, statistics and the formal aspects of computer and information science to support detailed analysis and modeling applicable to the discipline
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as professional responsibility of an engineer to public safety and sustainable development
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes

Range of Complex Engineering Problem Solving (Need High Taxonomy Level)

Attribute: Complex Engineering Problems have characteristics WP1 and some or all of WP2 to WP7

WP1	Depth of Knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamentals-based, first principles analytical approach
WP2	Range of conflicting requirements	Involve wide-ranging or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements
WP3	Depth of analysis required	Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models

WP4	Familiarity of issues	Involve infrequently encountered issues or novel problems
WP5	Extent of applicable codes	Address problems not encompassed by standards and codes of practice for professional engineering
WP6	Extent of stakeholder involvement and conflicting requirements	Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs
WP7	Interdependence	Address high level problems including many components or sub-problems that may require a systems approach

Range of Complex Engineering Activities (Project based)

Attributes: Complex activities mean (engineering) activities or projects that have some or all of the following characteristics:

EA1	Range of resources	Involve the use of diverse resources including people, data and information, natural, financial and physical resources and appropriate technologies including analytical and/or design software
EA2	Level of interaction	Require optimal resolution of interactions between wide-ranging and/or conflicting technical, non-technical, and engineering issues
EA3	Innovation	Involve creative use of engineering principles, innovative solutions for a conscious purpose, and research-based knowledge
EA4	Consequences to society and the environment	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
EA5	Familiarity	Can extend beyond previous experiences by applying principles-based approaches

Bloom's Taxonomy Domain:

Cognitive Domain		**Psychomotor Domain**		**Affective Domain**	
C1	Remembering	**P1**	Perception	**A1**	Receive
C2	Understanding	**P2**	Set	**A2**	Respond
C3	Applying	**P3**	Guided Response	**A3**	Value
C4	Analyzing	**P4**	Mechanism	**A4**	Organize
C5	Evaluating	**P5**	Complex Overt Response	**A5**	Internalize
C6	Creating/Designing	**P6**	Adaption		

Detail Lecture Plan

Module: 1

Lecture 1: Introduction to Ethics

Learning Outcomes:

1. Understand the fundamental concepts of ethics.
2. Analyze the role of ethics in engineering practices.

Topics Covered:

- Definition of ethics and its importance in professional life.
- The role of ethics in society and in professional fields.
- Ethical theories: Deontology, Utilitarianism, Virtue Ethics.

Case Study:

Ethical dilemmas in the development of AI systems.

Lecture 2: Theories of Ethics

Learning Outcomes:

1. Identify different ethical theories and their applications.
2. Critically assess ethical problems using ethical frameworks.

Topics Covered:

- Ethical theories overview: Consequentialism, Non-consequentialism, Virtue Ethics.
- Real-world applications of these theories in engineering.

Case Study:

Ethical implications of facial recognition technology in public spaces.

Lecture 3: Principles of Engineering Ethics

Learning Outcomes:

1. Understand the core principles of engineering ethics.
2. Apply these principles to real-life scenarios in engineering.

Topics Covered:

- Core principles: Safety, competence, integrity, confidentiality.
- Importance of ethics in engineering decision-making.

Case Study:

Ethics in designing software for medical devices.

Lecture 4: Employers and Employees in Engineering Ethics

Learning Outcomes:

1. Understand the ethical relationship between employers and employees.
2. Analyze professional responsibility in the workplace.

Topics Covered:

- Employer's responsibilities towards employees.
- Employee's responsibilities towards employers and clients.

Case Study:

Conflict of interest in the workplace in the tech industry.

Lecture 5: Inter-professional Relationship and Ethics

Learning Outcomes:

1. Recognize the importance of inter-professional ethics in engineering.
2. Apply ethical principles in collaborative work.

Topics Covered:

- Importance of teamwork in engineering projects.

- Ethics of collaboration across different engineering disciplines.

Case Study:
Ethical issues in cross-disciplinary collaboration in large tech projects.

Lecture 6: Fundamental Canons of Engineering Ethics

Learning Outcomes:

1. Understand the fundamental canons of engineering ethics.
2. Apply the canons in decision-making processes.

Topics Covered:

- NSPE (National Society of Professional Engineers) Canons.
- Key ethical principles that engineers must follow.

Case Study:
Violation of ethical canons in engineering projects.

Lecture 7: NSPE Codes and Ethical Conduct

Learning Outcomes:

1. Understand the significance of NSPE codes in professional ethics.
2. Explore the application of NSPE codes in real-world scenarios.

Topics Covered:

- Overview of NSPE code of ethics.
- Application in various engineering fields.

Case Study:
Ethical violations based on NSPE codes in civil engineering projects.

Lecture 8: IEEE Codes of Ethics

Learning Outcomes:

1. Learn about IEEE's ethical standards for engineers.
2. Analyze real-life cases of IEEE ethical dilemmas.

Topics Covered:

- IEEE Code of Ethics.
- Role of IEEE in promoting ethical practices in engineering.

Case Study:
Ethical issues surrounding the creation of an autonomous vehicle system.

Lecture 9: ACM Codes of Ethics

Learning Outcomes:

1. Understand the ACM code of ethics and its impact on computer science and engineering.
2. Critically assess cases based on ACM ethical standards.
3. **Topics Covered:**

- Overview of ACM (Association for Computing Machinery) codes.
- Relevance of ACM codes in software development and IT services.

Case Study:
Ethical dilemma in software piracy and copyright violations in software companies.

Lecture 10: Institutionalization of Ethical Conduct

Learning Outcomes:

1. Explore how ethical conduct becomes institutionalized in organizations.
2. Understand the role of ethics committees and professional bodies.

Topics Covered:

- Institutionalizing ethics within organizations.
- Ethical audits, ethics committees, and codes of conduct.

Case Study:
Case study on ethical audits in tech companies (e.g., Google's AI ethics board).

Lecture 11: Ethical Dilemmas in Engineering

Learning Outcomes:
1. Identify common ethical dilemmas faced by engineers.
2. Evaluate and resolve ethical dilemmas using ethical frameworks.

Topics Covered:
- Defining ethical dilemmas.
- Techniques for resolving ethical dilemmas in engineering.

Case Study:
Ethical dilemma involving a software company and user privacy concerns.

Lecture 12: Technology and Computer Ethics

Learning Outcomes:
1. Recognize the ethical challenges posed by emerging technologies.
2. Examine the role of computer science in addressing ethical issues.

Topics Covered:
- Ethical challenges in computer science and engineering.
- Key issues like algorithmic bias, AI ethics, and automation.

Case Study:
The ethical implications of using AI to replace human workers.

Lecture 13: Computer Crime and Cybersecurity Ethics

Learning Outcomes:
1. Understand the ethical implications of computer crimes.
2. Examine cybersecurity practices from an ethical standpoint.

Topics Covered:
- Cybersecurity threats and their ethical implications.
- Legal and ethical responsibilities of cybersecurity professionals.

Case Study:
Ethical challenges in the case of a data breach in a healthcare organization.

Lecture 14: Privacy and Confidentiality in CSE

Learning Outcomes:
1. Understand privacy and confidentiality issues in computer science.
2. Identify the ethical responsibilities of engineers in protecting user data.

Topics Covered:
- Privacy issues in digital technologies.
- Legal frameworks for privacy (GDPR, CCPA).

Case Study:
Ethical dilemma in sharing customer data by tech companies.

Lecture 15: Legal Framework for Computer Ethics

Learning Outcomes:
1. Learn about the legal aspects of ethics in computer science.
2. Understand the relationship between law and ethical practice in engineering.

Topics Covered:

- Overview of legal aspects of IT and computer science ethics.
- ICT Act, data protection laws.

Case Study:

Ethical and legal issues surrounding social media platforms' data handling.

Lecture 16: Copyright Laws and Ethics in CSE

Learning Outcomes:

1. Understand the importance of copyright laws in software development.
2. Explore ethical issues related to software piracy.

Topics Covered:

- Copyright, patents, and intellectual property rights.
- Ethical practices in software development and distribution.

Case Study:

Case of illegal software distribution and ethical violations in the tech industry.

Lecture 17: ICT Act and Ethical Responsibilities

Learning Outcomes:

1. Understand the role of the ICT Act in regulating ethical practices.
2. Analyze real-world applications of the ICT Act in CSE.

Topics Covered:

- Overview of ICT Act and its relevance to computer science ethics.
- Ethical implications of violating the ICT Act.

Case Study:

Case study on the misuse of the ICT Act in data breaches.

Lecture 18: Right to Information and Ethical Challenges

Learning Outcomes:

1. Examine the ethical implications of the Right to Information Act.
2. Understand the balance between transparency and privacy.

Topics Covered:

- Right to Information (RTI) Act and its impact on CSE professionals.
- Ethical challenges in managing public access to information.

Case Study:

Ethical issues in the release of personal data under RTI.

Lecture 19: Patents and Intellectual Property in CSE

Learning Outcomes:

1. Learn about patents and intellectual property in the tech industry.
2. Explore the ethical considerations in patenting software and technology.

Topics Covered:

- Patent laws and the role of intellectual property in technology.
- Ethical challenges in patenting software and algorithms.

Case Study:

Ethical dilemma in patenting AI-based algorithms.

Module: 2

Lecture 20: Ethical Challenges with Modern Technology in Society

Learning Outcomes:

1. Explore the ethical challenges of emerging technologies.
2. Identify the societal impact of technological advances.

Topics Covered:

- Ethical implications of AI, robotics, and automation.
- Societal impact of technological advances.

Case Study:

The ethical concerns surrounding autonomous weaponry.

Lecture 21: Responsible Professionalism in Engineering

Learning Outcomes:

1. Understand the responsibilities of a professional engineer.
2. Learn about ethical decision-making in professional contexts.

Topics Covered:

- Responsibilities of engineers as professionals.
- Ethical decision-making frameworks.

Case Study:

Ethical decisions in the development of autonomous vehicles.

Lecture 22: Moral Reasoning and Codes of Ethics

Learning Outcomes:

1. Learn about moral reasoning in professional ethics.
2. Understand how codes of ethics guide moral reasoning.

Topics Covered:

- The process of moral reasoning.
- How engineering codes of ethics guide professional behavior.

Case Study:

Moral reasoning in decisions regarding the safety of tech products.

Lecture 23: Ethical Decision-Making Strategies

Learning Outcomes:

1. Apply ethical decision-making strategies in engineering contexts.
2. Understand the role of personal and professional ethics in decision-making.

Topics Covered:

- Steps for ethical decision-making.
- Strategies for making ethical decisions in engineering projects.

Case Study:

Ethical decisions in the case of a critical software failure.

Lecture 24: Codes of Ethics and Case Studies

Learning Outcomes:

1. Explore various engineering codes of ethics.
2. Analyze real-world ethical dilemmas using case studies.

Topics Covered:

- Review of major engineering codes of ethics.
- Case studies from the engineering field.

Case Study:
Case study on ethical decisions in software engineering projects.

Lecture 25: Social Issues in Environmental Protection

Learning Outcomes:

1. Understand the social impact of environmental issues.
2. Identify the role of engineers in addressing environmental protection.

Topics Covered:

- Social dimensions of environmental issues.
- The role of engineers in promoting sustainability.

Case Study:
Social responsibility in the case of e-waste disposal.

Lecture 26: Sustainable Development Goals (SDGs)

Learning Outcomes:

1. Understand the concept of sustainable development.
2. Explore the SDGs and their relevance to engineering and technology.

Topics Covered:

- Overview of SDGs.
- Importance of SDGs in engineering practices.

Case Study:
Ethical implications of the tech industry's contribution to SDGs.

Lecture 27: Global Environmental Problems

Learning Outcomes:

1. Analyze the global environmental challenges we face.
2. Understand the role of engineering in solving environmental problems.

Topics Covered:

- Climate change, biodiversity loss, and pollution.
- Engineering solutions to global environmental issues.

Case Study:
The role of renewable energy in addressing environmental issues.

Lecture 28: Environmental Ethics and Pollution Prevention

Learning Outcomes:

1. Understand the concept of environmental ethics.
2. Explore strategies for pollution prevention in engineering.

Topics Covered:

- Principles of environmental ethics.
- Pollution prevention techniques in engineering.

Case Study:
Ethical considerations in the reduction of industrial emissions.

Lecture 29: Environmental Protection and Policy Tools

Learning Outcomes:

1. Understand the importance of environmental policy in sustainability.
2. Learn about environmental protection tools and techniques.

Topics Covered:

- Environmental policy tools: regulation, incentives, and penalties.
- Role of engineers in shaping environmental policies.

Case Study:
Ethical implications of environmental regulations in the tech industry.

Lecture 30: Waste Management and Solutions

Learning Outcomes:

1. Understand the ethical challenges in waste management.
2. Explore sustainable waste management practices.

Topics Covered:

- Environmental impact of waste.
- Engineering solutions for sustainable waste management.

Case Study:
E-waste management and its ethical implications.

Lecture 31: Climate Change and Industrialization

Learning Outcomes:

1. Explore the relationship between industrialization and climate change.
2. Understand the role of engineering in mitigating climate change.

Topics Covered:

- Impact of industrialization on climate change.
- Engineering innovations in reducing industrial carbon footprints.

Case Study:
Ethical issues in industrial contributions to climate change.

Lecture 32: Urban Environment and Sustainability

Learning Outcomes:

1. Examine the role of urbanization in sustainability.
2. Explore engineering solutions for sustainable urban environments.

Topics Covered:

- Urbanization and its environmental impact.
- Sustainable urban development and engineering solutions.

Case Study:

- Urban planning and ethical considerations in sustainable city designs.

Lecture 33: Voluntary Tools for Environmental Protection

Learning Outcomes:

1. Understand voluntary tools for environmental protection.
2. Analyze the role of voluntary initiatives in sustainability.

Topics Covered:

- Role of voluntary standards and certifications.
- Corporate social responsibility (CSR) and its impact on sustainability.

Case Study:
Corporate responsibility in reducing environmental impact.

Lecture 34: Environmental Legislation and Its Impact
Learning Outcomes: 1. Understand the role of environmental legislation in shaping ethical behavior. 2. Analyze the effectiveness of environmental laws. **Topics Covered:** • Importance of environmental legislation. • Evaluating the impact of environmental laws on engineering practices. **Case Study:** Case study on the effectiveness of environmental regulations in the automotive industry.

Lecture 35: Environmental Ethics and Social Responsibility
Learning Outcomes: 1. Explore the concept of social responsibility in engineering. 2. Understand the ethical considerations in sustainable development. **Topics Covered:** • Social responsibility and environmental ethics. • Engineering contributions to sustainable practices. **Case Study:** Ethical issues in tech companies' environmental responsibility.

Lecture 36: Sustainability Assessment Tools
Learning Outcomes: 1. Understand the role of sustainability assessment tools. 2. Apply sustainability tools in engineering projects. **Topics Covered:** • Sustainability assessment methods and tools. • Ethical and social responsibility in sustainability assessments. **Case Study:** Ethical decision-making in the application of sustainability tools in tech companies.

Approved By

Chairman, PSAC